

HUNDRED DAYS OF CODE

DAY 13

QUESTION: Write a menu-driven C program using switch-case to input a positive integer and count the number of even digits, and odd digits. **E.g.** Number: 12045, Even_dig = 3, Odd_dig = 2

INPUT: A positive integer

User's menu choice

For this option, the choice is 1 .

OUTPUT: Number of even digits

Number of odd digits

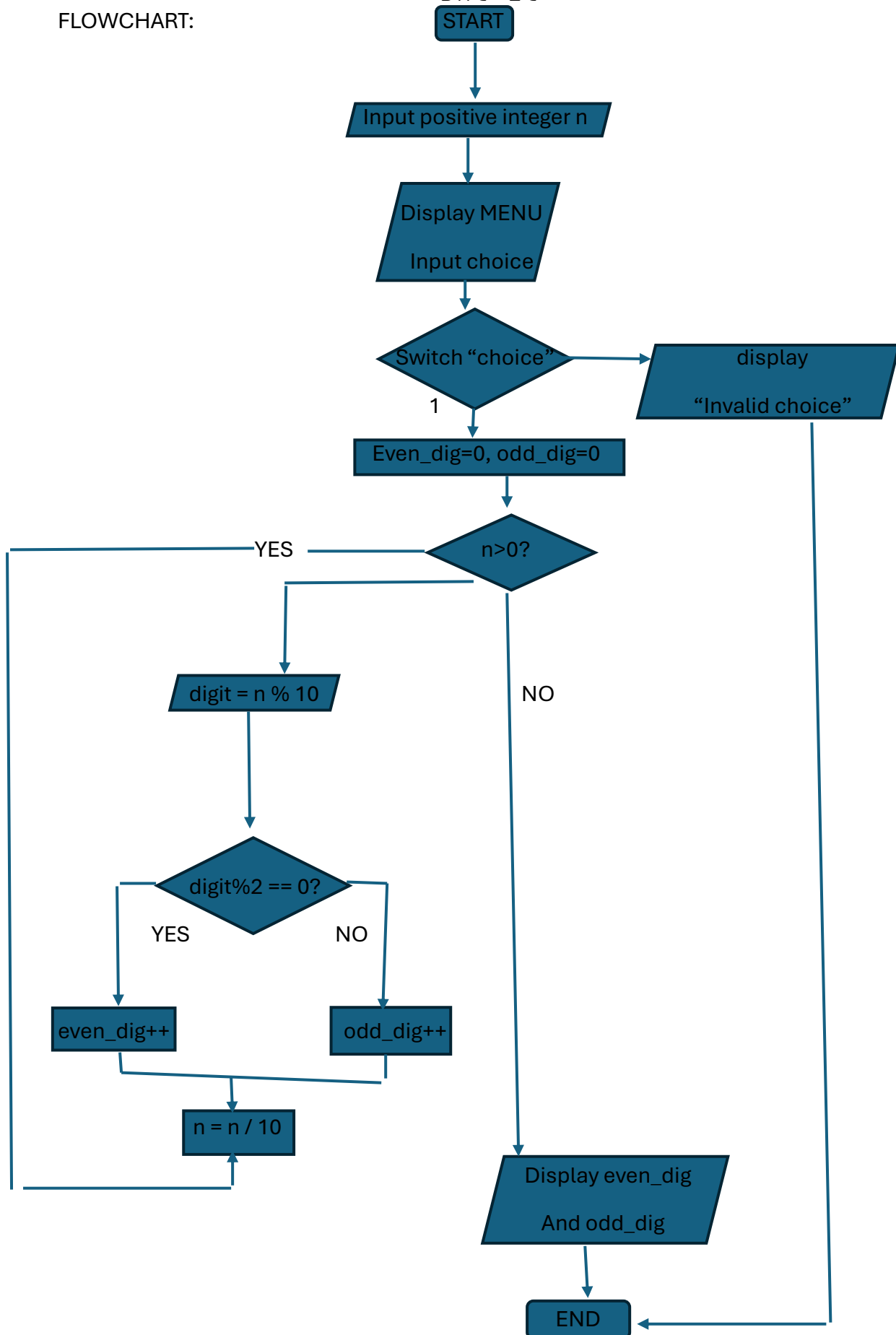
ALGORITHM:

1. Start.
2. Declare variables n, digit, choice, even_dig, and odd_dig.
3. Initialize even_dig = 0 and odd_dig = 0.
4. Input a positive integer n.
5. Display the menu.
6. Input the user's choice.
7. Use switch-case to select the operation.
8. If the choice is 1:
 - Repeat while n > 0.
 - Extract the last digit using digit = n % 10.
 - If digit % 2 == 0, increment even_dig.
 - Otherwise, increment odd_dig.
 - Remove the last digit using n = n / 10.
9. Display even_dig and odd_dig.
10. Stop.

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FLOWCHART:



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PSEUDOCODE: START

DECLARE n, choice, digit

DECLARE even_dig, odd_dig

even_dig $\leftarrow$ 0

odd_dig $\leftarrow$ 0

INPUT n

DISPLAY "1. Count even and odd digits"

INPUT choice

SWITCH choice

CASE 1:

WHILE n > 0

digit $\leftarrow$ n MOD 10

IF digit MOD 2 = 0 THEN

even_dig $\leftarrow$ even_dig + 1

ELSE

odd_dig $\leftarrow$ odd_dig + 1

END IF

n $\leftarrow$ n / 10

END WHILE

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DISPLAY even_dig

DISPLAY odd_dig

BREAK

DEFAULT:

DISPLAY "Invalid choice"

END SWITCH

STOP

TEST CASE TABLE:

TEST CASE	INPUT NUMBER	CHOICE	EXPECTED EVEN AND ODD DIGITS	ACTUAL OUTPUT	RESULT
1	12045	1	3,2	3,2	PASS
2	24680	1	5,0	5,0	PASS
3	12345	1	--,--	INVALID CHOICE	PASS

REQUIRED C PROGRAM:

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int choice, n, digit;
```

```
    int even_dig = 0, odd_dig = 0;
```

```
    printf("Enter a positive integer: ");
```

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```
scanf("%d", &n);

printf("\nMENU\n");
printf("1. Count even and odd digits\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        while (n > 0)
        {
            digit = n % 10;

            if (digit % 2 == 0)
                even_dig++;
            else
                odd_dig++;

            n = n / 10;
        }

        printf("Even_dig = %d\n", even_dig);
        printf("Odd_dig = %d\n", odd_dig);
        break;

    default:
        printf("Invalid choice.\n");
```

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```
}
```

```
return 0;
```

```
}
```

COMPILATION AND EXECUTION:

```
(kali㉿kali)-[~]  
└─$ cc digit.c -o digit  
  
(kali㉿kali)-[~]  
└─$ ./digit  
Enter a positive integer: 12045  
  
MENU  
1. Count even and odd digits  
Enter your choice: 1  
Even_dig = 3  
Odd_dig = 2  
  
(kali㉿kali)-[~]  
└─$ ./digit  
Enter a positive integer: 24680  
  
MENU  
1. Count even and odd digits  
Enter your choice: 1  
Even_dig = 5  
Odd_dig = 0  
  
(kali㉿kali)-[~]  
└─$ ./digit  
Enter a positive integer: 12345  
  
MENU  
1. Count even and odd digits  
Enter your choice: 2  
Invalid choice.  
  
(kali㉿kali)-[~]  
└─$
```

.....